

Bayesian Double Machine Learning for Causal Inference

Francis J. DiTraglia¹ Laura Liu²

¹University of Oxford

²University of Pittsburgh

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The Problem / Model

$$Y_i = \alpha D_i + X_i' \beta + \varepsilon_i, \quad \mathbb{E}[\varepsilon | D_i, X_i] = 0, \quad i = 1, \dots, n$$

Selection-on-observables

Learn causal effect α of D_i on Y_i ; treatment “as good as random” given p controls X_i .

Many Controls

Adjust for many covariates to make selection-on-observables plausible: p is large.

Bias-Variance Tradeoff

- ▶ OLS: unbiased but noisy when p large relative to n ; doesn't exist when $p > n$
- ▶ Drop control $X^{(j)}$ correlated with $D \Rightarrow$ biased estimate of α if $\beta^{(j)} \neq 0$

Example: Abortion and Crime

Donohue III & Levitt (2001; QJE); Belloni, Chernozhukov & Hansen (2014; ReStud)

Data: 48 states $\times$ 12 years ($n = 576$)

- ▶ Y_{it} : Crime rate (violent / property / murder)
- ▶ D_{it} : Effective abortion rate

D&L Controls

State fixed effects, time trends, 8 time-varying state controls

BCH Controls

Add quadratics, interactions, initial conditions $\times$ trends $\Rightarrow p/n \approx 0.5$

This Paper

- ▶ Bayesian causal inference with many controls.
- ▶ Don't select variables; *shrink* their coefficients (more stable than LASSO).
- ▶ Naïve Bayesian approach can be highly biased.
- ▶ Re-parameterization solves the problem: simple, fully-Bayesian inference.
- ▶ Match asymptotic properties of (Frequentist) Double Machine Learning methods.
- ▶ Better finite-sample performance: lower bias, better coverage, lower RMSE.

Start by explaining why “naïve” approach doesn't work.

Naïve Shrinkage: Ridge Regression (centered / scaled)

Minimize $(Y - \alpha D - X\beta)'(Y - \alpha D - X\beta) + \tau\beta'\beta$

$$\hat{\alpha}_\tau = \frac{D'M_\tau Y}{D'M_\tau D}, \quad M_\tau \equiv \mathbb{I}_n - X(X'X + \tau\mathbb{I}_p)^{-1}X' \quad (\text{Note: } M_\tau X \neq 0)$$

Compare with OLS (FWL Theorem)

$$\hat{\alpha}_{\text{OLS}} = \frac{(M_X D)'(M_X Y)}{(M_X D)'(M_X D)} = \frac{D'M_X Y}{D'M_X D}, \quad M_X \equiv \mathbb{I}_n - X(X'X)^{-1}X'$$

Bayesian Interpretation

Posterior mean: known σ_ε^2 , flat prior on α , iid $\text{Normal}(0, \sigma_\varepsilon^2/\tau)$ prior on β_j

Bias of Naïve Approach – Regularization-Induced Confounding (RIC)

Ridge: (analogous for LASSO etc. but no simple expression)

$$\hat{\alpha}_\tau = \frac{D' M_\tau Y}{D' M_\tau D} = \frac{D' M_\tau (\alpha D + X\beta + \varepsilon)}{D' M_\tau D} = \alpha + \underbrace{\frac{D' M_\tau X\beta}{D' M_\tau D}}_{\text{bias}} + \underbrace{\frac{D' M_\tau \varepsilon}{D' M_\tau D}}_{\text{mean-zero noise}}$$

Moment Condition for α evaluated at *true* β versus $\tilde{\beta} \neq \beta$

$$\mathbb{E}[\varepsilon D] = \mathbb{E}[(Y - X'\beta - \alpha D)D] = 0 \iff \alpha = \frac{\mathbb{E}[(Y - X'\beta)D]}{\mathbb{E}[D^2]}$$

$$\tilde{\alpha} = \frac{\mathbb{E}[(Y - X'\tilde{\beta})D]}{\mathbb{E}[D^2]} = \frac{\mathbb{E}[(Y - X'\beta)D + X'(\beta - \tilde{\beta})D]}{\mathbb{E}[D^2]} = \alpha + (\beta - \tilde{\beta})' \frac{\mathbb{E}[XD]}{\mathbb{E}[D^2]}$$

Adding a “first-stage” doesn’t help.

$$Y = \alpha D + X'\beta + \varepsilon, \quad \mathbb{E}[\varepsilon|X, D] = 0; \quad D = X'\gamma + V, \quad \mathbb{E}[VX] = 0$$

Implication

$$\text{Cov}(\varepsilon, V) = \text{Cov}(\varepsilon, D - X'\gamma) = \text{Cov}(\varepsilon, D) - \text{Cov}(\varepsilon, X')\gamma = 0.$$

+ Gaussian (quasi-)likelihood $\Rightarrow \varepsilon \perp\!\!\!\perp V$

$$\begin{aligned} \pi(\theta|Y, D, X) &\propto f(Y, D|X, \theta) \times \pi(\theta) \\ &= f(Y|D, X, \alpha, \beta, \sigma_\varepsilon^2) \times f(D|X, \gamma, \sigma_V^2) \times \pi(\alpha, \beta, \sigma_\varepsilon^2, \gamma, \sigma_V^2) \end{aligned}$$

Unless prior treats β and γ as *dependent*, adding the D on X regression has *no effect*!

Solution: Replace Structural Equation with *Another* Reduced Form

$$Y = \alpha D + X'\beta + \varepsilon, \quad \mathbb{E}[\varepsilon|X, D] = 0$$

$$D = X'\gamma + V, \quad \mathbb{E}[VX] = 0$$

Substitute for D

$$Y = \alpha D + X'\beta + \varepsilon = X'(\alpha\gamma + \beta) + (\varepsilon + \alpha V) = X'\delta + U$$

Backing out α

$$\text{Cov}(U, V) = \text{Cov}(\varepsilon + \alpha V, V) = \alpha \text{Var}(V) \quad \implies \quad \alpha = \frac{\text{Cov}(U, V)}{\text{Var}(V)} = \frac{\mathbb{E}[UV]}{\mathbb{E}[V^2]}$$

Our Approach: Bayesian Double Machine Learning (BDML)

$$Y_i = \alpha D_i + X_i' \beta + \varepsilon_i = X_i'(\alpha \gamma + \beta) + (\varepsilon_i + \alpha V_i) = X_i' \delta + U_i$$

$$\begin{aligned} Y_i &= X_i' \delta + U_i \\ D_i &= X_i' \gamma + V_i \end{aligned} \quad \left[\begin{array}{c} U_i \\ V_i \end{array} \right] \bigg| X_i \sim \text{Normal}_2(0, \Sigma), \quad \Sigma = \begin{bmatrix} \sigma_\varepsilon^2 + \alpha^2 \sigma_V^2 & \alpha \sigma_V^2 \\ \alpha \sigma_V^2 & \sigma_V^2 \end{bmatrix}$$

BDML Algorithm

1. Gaussian (quasi-)likelihood
2. Place “standard” priors on reduced form parameters (δ, γ, Σ)
3. Draw from posterior $(\delta, \gamma, \Sigma) | (X, D, Y)$
4. Posterior draws for $\Sigma \implies$ posterior draws for $\alpha = \sigma_{UV} / \sigma_V^2$

Why “Double” Helps: small $\times$ small = smaller

Naïve

$$\mathbb{E}[(Y - X'\tilde{\beta} - \tilde{\alpha}D)D] = 0 \iff \tilde{\alpha} = \alpha + (\beta - \tilde{\beta})' \frac{\mathbb{E}[XD]}{\mathbb{E}[D^2]}$$

Double

$$\mathbb{E}[(\hat{U} - \hat{\alpha}\hat{V})\hat{V}] = \mathbb{E}\left[\left\{(Y - X'\hat{\delta}) - \hat{\alpha}(D - X'\hat{\gamma})\right\}(D - X'\hat{\gamma})\right] = 0 \iff \hat{\alpha} = \frac{\mathbb{E}[\hat{U}\hat{V}]}{\mathbb{E}[\hat{V}^2]}$$

$$\mathbb{E}[\hat{U}\hat{V}] = \mathbb{E}\left[\left\{U + X'(\delta - \hat{\delta})\right\}\left\{V + X'(\gamma - \hat{\gamma})\right\}\right] = \mathbb{E}[UV] + (\delta - \hat{\delta})\mathbb{E}[XX'](\gamma - \hat{\gamma})$$

$$\mathbb{E}[\hat{V}^2] = \mathbb{E}\left[\left\{V + X'(\gamma - \hat{\gamma})\right\}^2\right] = \mathbb{E}[V^2] + (\gamma - \hat{\gamma})'\mathbb{E}[XX'](\gamma - \hat{\gamma})$$

Why “Double” Helps: doesn’t assume away selection bias!

$$\text{Selection Bias} \equiv \frac{\text{Cov}(Y, D)}{\text{Var}(D)} - \alpha = \frac{\beta' \mathbb{E}[XX'] \gamma}{\sigma_V^2 + \gamma' \mathbb{E}[XX'] \gamma}$$

Sims (2012)

Reasonable low-dimensional priors “can unintentionally imply dogmatic beliefs about parameters of interest” when expanded “unthinkingly to high dimensions.”

Naïve

If $\gamma \perp\!\!\!\perp \beta$, implied prior for Selection Bias is a **point mass at zero** for p large.

BDML

If $\gamma \perp\!\!\!\perp \delta$, implied prior for Selection Bias **centered at zero but non-degenerate** for large p .

BDML versus Frequentist Double Machine Learning (FDML)

FDML Optimizes

Plug in “Machine Learning” estimators of reduced form parameters: $(\hat{\delta}_{\text{ML}}, \hat{\gamma}_{\text{ML}})$

$$\hat{\alpha}_{\text{FDML}} = \frac{\sum_{i=1}^n (Y_i - X_i' \hat{\delta}_{\text{ML}})(D_i - X_i' \hat{\gamma}_{\text{ML}})}{\sum_{i=1}^n (D_i - X_i' \hat{\gamma}_{\text{ML}})^2}.$$

Finite-Sample Concerns

Wüthrich & Zhu (2023), Bach et al. (2024), Ahrens et al. (2025)

BDML Marginalizes

Posterior for α averages over uncertainty about γ and δ and applies shrinkage to Σ .

Theoretical Results

$$\pi(\Sigma, \delta, \gamma) \propto \pi(\Sigma)\pi(\delta)\pi(\gamma)$$

$$\begin{aligned} Y_i &= X_i' \delta + U_i \\ D_i &= X_i' \gamma + V_i \end{aligned} \quad \left[\begin{array}{c} U_i \\ V_i \end{array} \right] \bigg| X_i \sim \text{Normal}_2(0, \Sigma)$$

$$\Sigma \sim \text{Inverse-Wishart}(\nu_0, \Sigma_0)$$

$$\delta \sim \text{Normal}_p(0, \mathbb{I}_p / \tau_\delta)$$

$$\gamma \sim \text{Normal}_p(0, \mathbb{I}_p / \tau_\gamma)$$

Naïve Approach

Analogous but with single structural equation and $\beta \sim \text{Normal}(0, \mathbb{I}_p / \tau_\beta)$

Asymptotic Framework

Fixed true parameters $(\Sigma^*, \delta^*, \gamma^*)$; $n \rightarrow \infty$ (large sample); $p \rightarrow \infty$ (many controls)

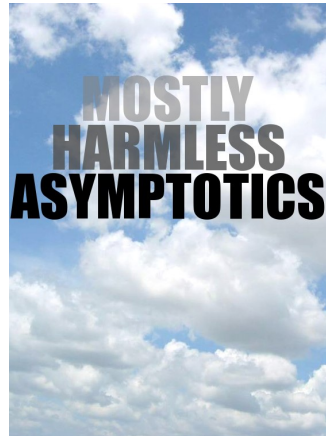
Our asymptotic framework ensures bounded R-squared.

Rate Restrictions

- (i) sample size dominates # of controls: $p/n \rightarrow 0$
- (ii) sample size dominates prior precisions: $\tau/n \rightarrow 0$
- (iii) precisions of same order as # controls: $\tau \asymp p$

Regularity Conditions

- (i) $p < n$
- (ii) $\text{Var}(X) \equiv \Sigma_X$ “well-behaved” as $p \rightarrow \infty$
- (iii) $\lim_{p \rightarrow \infty} \sum_{j=1}^p (\beta_j^*)^2 < \infty$, $\lim_{p \rightarrow \infty} \sum_{j=1}^p (\gamma_j^*)^2 < \infty$
- (iv) iid errors/controls, $\mathbb{E}(X_i) = 0$, finite & p.d. Σ^*



Asymptotic Results: Bias and Consistency

Consistency and Bias

All three estimators are consistent with the same asymptotic variance if $p/\sqrt{n} \rightarrow 0$.

- ▶ Naïve: bias of order p/n
- ▶ BDML and FDML: bias of order $(p/n)^2$

$\sqrt{n}$ -Consistency

- ▶ Naïve requires $p/\sqrt{n} \rightarrow 0$
- ▶ BDML and FDML require only $p/n^{3/4} \rightarrow 0$

Asymptotic Results: Bernstein-von Mises

Bernstein-von Mises Theorem for BDML

- ▶ BDML posterior for α : asymptotically normal, correct Frequentist coverage
- ▶ Credible intervals are valid confidence intervals
- ▶ Semiparametrically efficient

Comparison with Existing Results

- ▶ Builds on Walker (2025); we extend to sub-Gaussian X_i and empirical L_2 -norm
- ▶ Weaker assumptions than Luo et al. (2023), Breunig et al. (2024)
- ▶ Robust to misspecification of error distribution

Simulation Experiment

Baseline: $n = 200$, $p = 100$, $\alpha = 1/4$, $R_D^2 = R_Y^2 = 0.5$; vary ρ

$$Y_i = \alpha D_i + X_i' \beta + \varepsilon_i \quad X_i \sim \text{Normal}_p(0, \mathbb{I}_p)$$

$$D_i = X_i' \gamma + V_i \quad (\varepsilon_i, V_i) \sim \text{Normal}_2 \left(0, \text{diag}\{1 - R_Y^2, 1 - R_D^2\} \right)$$

$$(\beta_j, \gamma_j)' \sim \text{Normal} \left(\mathbf{0}, \frac{1}{p} \begin{pmatrix} R_Y^2 & \rho \sqrt{R_Y^2 R_D^2} \\ \rho \sqrt{R_Y^2 R_D^2} & R_D^2 \end{pmatrix} \right)$$

- ▶ R_D^2, R_Y^2 : how well X predicts D and Y (partial)
- ▶ $\rho \equiv \text{Corr}(\beta_j, \gamma_j)$; Selection bias = $\rho \sqrt{R_D^2 R_Y^2}$

BDML Prior Specifications

BDML-IW (Theory)

- ▶ $\Sigma \sim \text{Inverse-Wishart}(4, I_2)$
- ▶ $\delta \sim \text{Normal}_p(0, \mathbb{I}_p/\tau_\delta)$, $\gamma \sim \text{Normal}_p(0, \mathbb{I}_p/\tau_\gamma)$, with $\tau_\delta, \tau_\gamma \asymp p$

BDML-LKJ-HP (Practice)

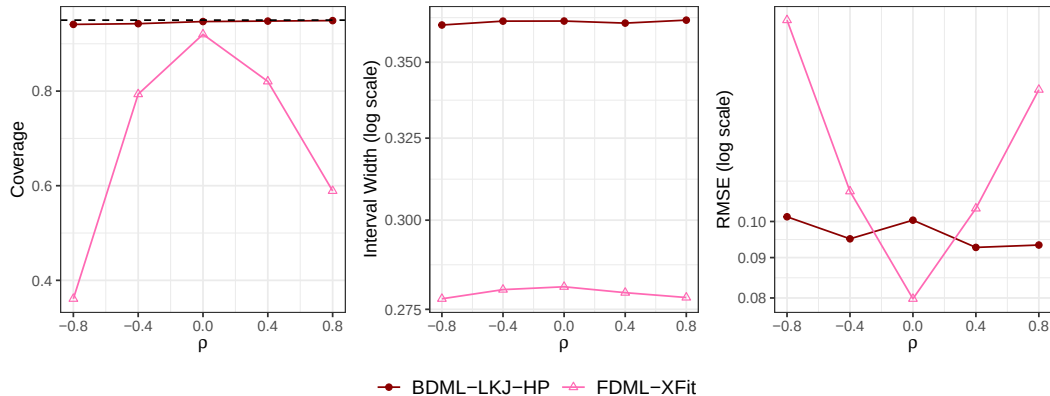
- ▶ Σ : LKJ(4) on $\text{Corr}(U, V)$; $\text{Cauchy}^+(0, 2.5)$ on SDs
- ▶ (δ, γ) : $\text{Normal}(0, \sigma^2 I)$ with $\sigma^2 \sim \text{Inv-Gamma}(2, 2)$

BDML is pretty robust

We've tried a number of alternative priors; they give similar results.

Simulation Results: BDML vs FDML

Baseline: $R_D^2 = R_Y^2 = 0.5$, $\alpha = 1/4$, $n = 200$, $p = 100$



Two-Step “Plug-in” Bayesian Approaches

Preliminary Regression

$\hat{D}_i \equiv X_i' \hat{\gamma}_{\text{prelim}} \leftarrow$ estimate from Bayesian regression of D on X .

HCPH (Hahn et al, 2018; Bayesian Analysis)

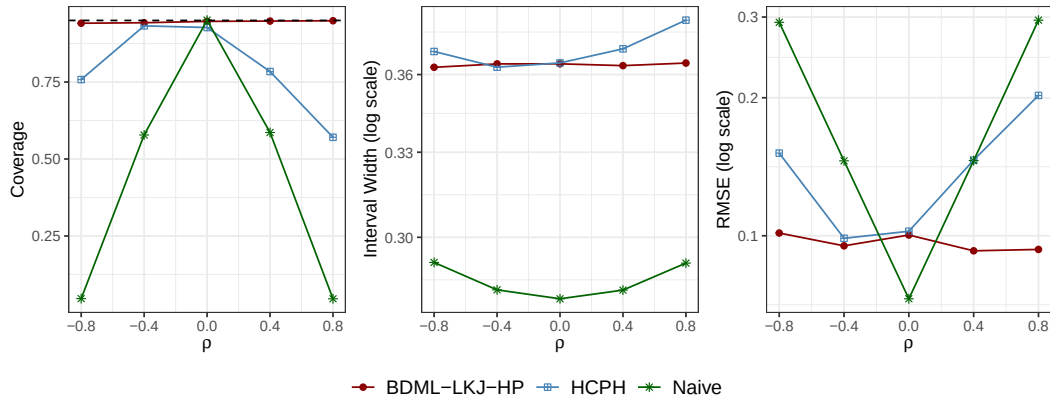
1. Bayesian linear regression of Y on $(D - \hat{D})$ and X
2. Estimation / inference for α from posterior for $(D - \hat{D})$ coefficient.

Linero (2023; JASA)

1. Bayesian linear regression of Y on $(D, \hat{D}, X)$.
2. Estimation / inference for α from posterior for the D coefficient.

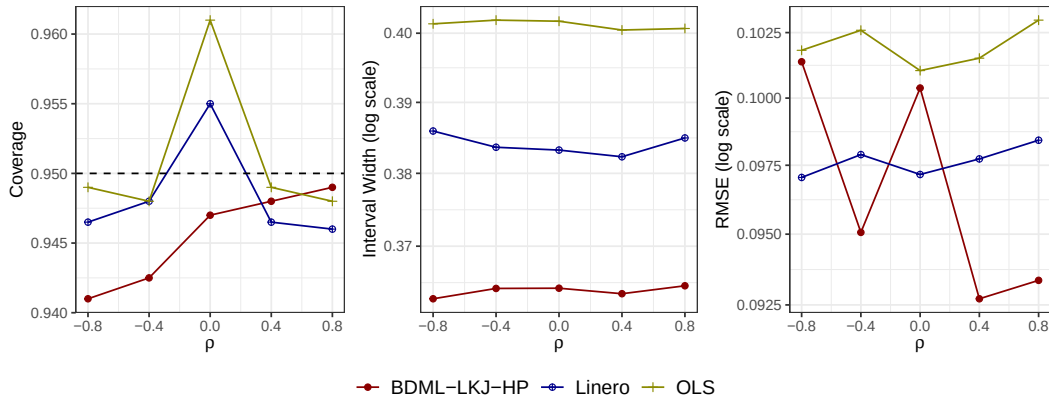
Simulation Results: BDML vs HCPH, Naïve

Baseline: $R_D^2 = R_Y^2 = 0.5$, $\alpha = 1/4$, $n = 200$, $p = 100$



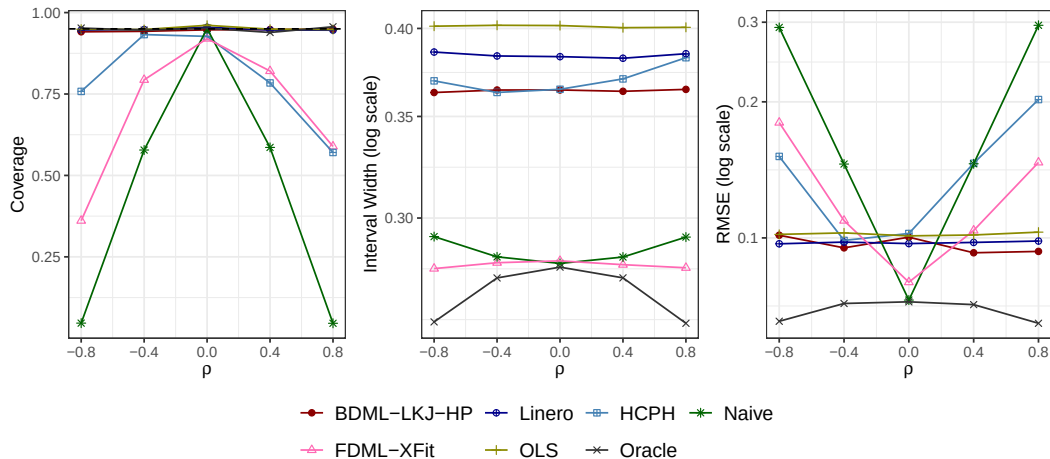
Simulation Results: BDML vs Linero, OLS

Baseline: $R_D^2 = R_Y^2 = 0.5$, $\alpha = 1/4$, $n = 200$, $p = 100$



Simulation Results: All Estimators

Baseline: $R_D^2 = R_Y^2 = 0.5$, $\alpha = 1/4$, $n = 200$, $p = 100$

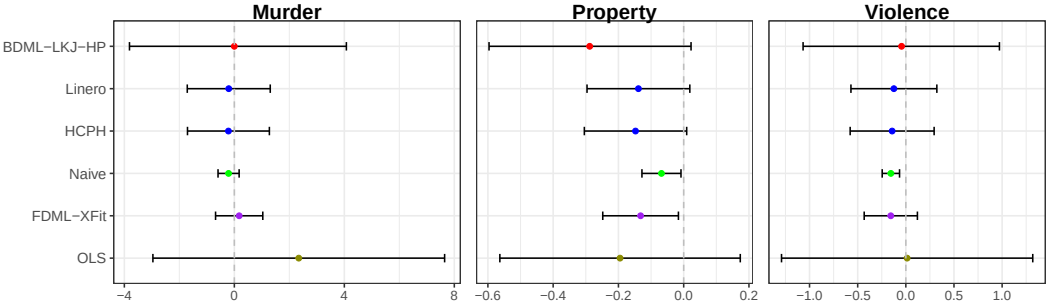


Example: Effect of Abortion on Crime

- ▶ Recall: Donohue III & Levitt (2001) as revisited by BCH (2014)
- ▶ ΔY_{it} : change in crime rate; ΔD_{it} : change in effective abortion rate
- ▶ X_{it} : baseline controls, lags, squared lags, state-level controls $\times$ trends

Outcome	n	p	R_D^2	R_Y^2
Murder	576	282	0.99	0.41
Property	576	282	0.99	0.59
Violence	576	282	1.00	0.59

Levitt Results: Effect of Abortion on Crime



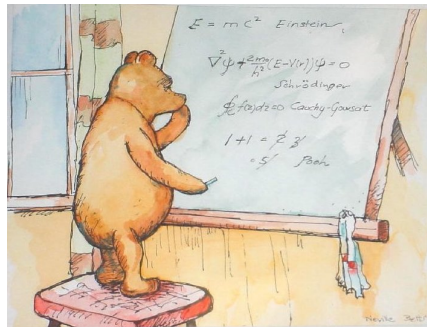
Thanks for listening!

Summary

- ▶ Simple, fully-Bayesian causal inference in a workhorse linear model with many controls.
- ▶ Avoids RIC; Excellent Frequentist Properties

In Progress

- ▶ Extensions: partially linear model; treatment interactions; instrumental variables.



Appendix

Point Estimates of α

	Naïve		B/FDML
Asymp. bias	p/n	$\succ$	p^2/n^2
Asymp. s.e.	$1/\sqrt{n}$	$\sim$	$1/\sqrt{n}$
Consistency	✓		✓
$\sqrt{n}$ -consistency	if $p = o(\sqrt{n})$		if $p = o(n^{3/4})$

Details: [▶ Naïve](#) [▶ BDML](#) [▶ FDML](#) [▶ Variances](#)

- ▶ Under standard regularity conditions with $\text{Var}(D_i) = \text{Var}(Y_i) = 1$ and $p \succ \sqrt{n}$, let $R = \delta^* \Sigma_X \gamma^*$ capture confounding,

$$\partial \text{AsympBias}_{\text{BDML}} / \partial R \leq \partial \text{AsympBias}_{\text{FDML}} / \partial R$$

Posterior Distribution of α : BvM Theorem

Proposition (Bernstein-von Mises theorem)

Assume the previous setup and assumptions hold, except that σ_ϵ^{2*} and σ_V^{2*} are known, the prior for α is continuous and positive over a neighborhood of α^* , and α^* is in the interior of its support. Further suppose that X_i is sub-Gaussian, and the eigenvalues of Σ_X are uniformly bounded, $p = o(\sqrt{n})$, and $\log n = o(p^{1/2})$. Then, as $n \rightarrow \infty$,

$$\left\| P(\alpha \in \cdot | X, W) - \mathcal{N}\left(\alpha^* + \frac{\tilde{\Delta}_n^*}{\sqrt{n}}, \frac{1}{n} I_n(\gamma^*)^{-1}\right) \right\|_{TV} \xrightarrow{P} 0,$$

where $\tilde{I}_n(\gamma) = \frac{1}{n} \sum_{i=1}^n \frac{(D_i - X_i' \gamma)^2}{\sigma_\epsilon^{2*}}$ and

$$\tilde{\Delta}_n^* = \tilde{I}_n(\gamma^*)^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{[Y_i - \alpha^* (D_i - X_i' \gamma^*) - X_i' \delta^*] (D_i - X_i' \gamma^*)}{\sigma_\epsilon^{2*}}.$$

Point Estimates of α

Proposition (Naïve)

Assume the previous setup and assumptions hold. Then, the posterior mean of α is given by

$$\hat{\alpha}_{naive} = \alpha^* + \frac{\lambda}{n} \frac{\gamma^{*'} \beta^*}{\Sigma_{22}^*} + o\left(\frac{p}{n}\right) + O_p\left(\frac{1}{\sqrt{n}}\right),$$

where the $O_p(1/\sqrt{n})$ term is mean zero. Therefore, the naive estimator is consistent, but not $\sqrt{n}$ -consistent for $\lim_{n \rightarrow \infty} p^2/n = \infty$ and $\gamma^{'} \beta^* \neq 0$.*

► back

Point Estimates of α

Proposition (BDML)

Assume the previous setup and assumptions hold. The posterior mean of $\alpha|X, W$ under the BDML estimator is given by

$$\begin{aligned}\hat{\alpha}_{BDML} &= \mathbb{E}[\alpha|X, W] \\ &= \alpha^* + \frac{\alpha^*}{n} \left(\frac{\Sigma_{0,12} + (\Sigma^* C^* \Sigma^*)_{12}}{\Sigma_{12}^*} - \frac{\Sigma_{0,22} + (\Sigma^* C^* \Sigma^*)_{22}}{\Sigma_{22}^*} \right) \\ &\quad + o_p\left(\frac{1}{n}\right) + o_p\left(\left(\frac{p}{n}\right)^2\right) + O_p\left(\frac{1}{\sqrt{n}}\right),\end{aligned}$$

where

$$C^* = \frac{1}{n} \begin{bmatrix} \tau_\delta^2 \delta^{*'} \Sigma_X^{-1} \delta^* - \tau_\delta \text{tr}(\Sigma_X^{-1}) & \tau_\delta \tau_\gamma \delta^{*'} \Sigma_X^{-1} \gamma^* \\ \tau_\delta \tau_\gamma \gamma^{*'} \Sigma_X^{-1} \delta^* & \tau_\gamma^2 \gamma^{*'} \Sigma_X^{-1} \gamma^* - \tau_\gamma \text{tr}(\Sigma_X^{-1}) \end{bmatrix},$$

and the $O_p\left(\frac{1}{\sqrt{n}}\right)$ term is mean zero. Therefore, $\hat{\alpha}_{BDML}$ is consistent. Moreover, if $p = o(n^{3/4})$, $\hat{\alpha}_{BDML}$ is $\sqrt{n}$ -consistent.

Point Estimates of α

Remark (FDML)

Assume the previous setup and assumptions hold. For the FDML estimator based on the full sample,

$$\hat{\alpha}_{FDML} = \alpha^* + \frac{\alpha^*}{n} \left(\frac{(C_2^*)_{12}}{\Sigma_{12}^*} - \frac{(C_2^*)_{22}}{\Sigma_{22}^*} \right) + o_p \left(\frac{1}{n} \right) + o_p \left(\left(\frac{p}{n} \right)^2 \right) + O_p \left(\frac{1}{\sqrt{n}} \right),$$

where

$$C_2^* = \frac{1}{n} \begin{bmatrix} \tau_\delta^2 \delta^{*'} \Sigma_X^{-1} \delta^* & \tau_\delta \tau_\gamma \delta^{*'} \Sigma_X^{-1} \gamma^* \\ \tau_\delta \tau_\gamma \gamma^{*'} \Sigma_X^{-1} \delta^* & \tau_\gamma^2 \gamma^{*'} \Sigma_X^{-1} \gamma^* \end{bmatrix},$$

and the $O_p \left(\frac{1}{\sqrt{n}} \right)$ term is mean zero. Therefore, $\hat{\alpha}_{FDML}$ is consistent. Moreover, if $p = o(n^{3/4})$, $\hat{\alpha}_{FDML}$ is $\sqrt{n}$ -consistent.

Point Estimates of α

Proposition (Variances of $\hat{\alpha}$)

Assume the previous setup and assumptions hold. The asymptotic variance of $\hat{\alpha}_{naive}$, $\hat{\alpha}_{BDML}$ and $\hat{\alpha}_{FDML}$ are the same, and are given by $\frac{\Sigma_{11}^ \Sigma_{22}^* - (\Sigma_{12}^*)^2}{(\Sigma_{22}^*)^2}$.*

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